

Canonical source: 00-CHARTER/GUIDES/FEATURES_GUIDE.md

Æ Orange AI Computer Features Guide

This guide explains the Æ Orange AI Computer capability map. It does not promote any feature. Status must be refreshed from live probes and receipts.

The accepted 2026-08-28 integrated snapshot is green at `10-RECEIPTS/orange5-build/2026-08-28T03-42-45-242Z-integrated-operational-proof.json`. Blue Bench accepted 10/10 exact-path lanes in `10-RECEIPTS/orange5-build/2026-08-28T03-40-44-768Z-blue-bench.json`. The later Hermes Brain MCP delegation also completed with all ten checks true in `10-RECEIPTS/orange5-build/2026-08-28T04-13-22-203Z-brain-mcp-delegation-live-proof.json`.

Status Vocabulary

- `PROVEN_SNAPSHOT` : a referenced historical or recent receipt proved the exact path at that time. Re-probe before relying on it now.
- `CONFIGURED_UNPROVEN` : source and configuration exist; live proof is pending.
- `DEGRADED_OPERATIONAL` : callable with a named limitation.
- `BLOCKED` : required runtime, authority, or evidence is missing.
- `FUTURE_CANDIDATE` : research or planned lane; not active.

1. Core Control Features

Feature	What it does	Evidence posture
Bun Navigator Kernel	Zero-model-RAM reflex for schemas, policy, health, routing, and repeated work	Current runtime implementation; verify live health
Canonical Spine	Runs scoped orders through gates, memory, compression, models, effectors, and receipts	Current source and focused proofs exist; verify current run
orange.order.v1	Stable operational input contract	Implemented and validated by CLI/MCP
orange.report.v1	Stable operational output contract	Deterministically compiled; model has no receipt authority
LOOM	Authority, scope, risk, evidence, approval, and false-green gates	Current Hermes/Spine code; exact action still needs a receipt
Project Lock	Keeps orders attached to the declared target and scope	Current control feature; verify on the active project

Feature	What it does	Evidence posture
Project Continuum	Preserves transcripts, sources, lineage, and duplicate-work preflight	Current memory path; quality is task-dependent
Hash-chained receipts	Records actual actions and evidence with lineage	Core operational truth layer

2. OrangeBrain And Client Features

Feature	What it does	Evidence posture
OpenAI-compatible gateway	Lets Atomic Orange and compatible clients use governed model routes	Endpoint configured at loopback; probe <code>/healthz</code>
Three-speed routing	Chooses Bun reflex, Navigator, or specialist	Current source; route receipt proves actual choice
Model output repair	Converts compact model output to public report schema and rejects false authority	Implemented in gateway/control path
Brain MCP stdio	Desktop and IDE integration	Current dual-transport proof green with 10 tools
Brain MCP HTTP	Loopback app and model-to-model integration	Current authenticated loopback proof green with 12 tools and protocol/header/body guards
Durable MCP Tasks	Long tool calls for current protocol clients	Implemented in task store; live client support varies
Atomic Orange	Optional native operator interface	Interface lane; not required for headless operation

3. Memory And Learning Features

Feature	What it does	Evidence posture
AE Memory / Cobra	Durable facts, decisions, failures, and retrieval	Current held-out quality: 23/23, hybrid MRR <code>0.9058</code> , p95 <code>445 ms</code>
Lexical retrieval	Exact keyword and identifier recall	Implemented
Dense retrieval	Semantic recall through local embedding lane	Configured; active model must be proved
Hybrid retrieval	Combines lexical and dense scores	Prior benchmark evidence exists; rerun for current data
Failure Memory	Surfaces prior matching failures before action	Current source and focused proof path

Feature	What it does	Evidence posture
Contradiction debt	Records incompatible claims instead of merging them	Current memory implementation
Receipt-to-Reflex	Promotes repeated proven work into deterministic behavior	Governed learning path; each rule needs falsification and rollback
Superdirectory	Stores project docs and transcripts on disk for bounded hydration	Implemented maintenance commands

4. Compression Features

Canonical engine: `12-ATOMSMASHER/full-scope` .

Feature	What it does
Commitment Atoms	Preserves irreducible laws, decisions, constraints, and tasks
AIR Codec	Turns verbose text into typed structural frames
Equation Memory	Stores numeric rules plus residuals where justified
Cartridges	Reuses stable domain context without rebuilding it
Sparse Worksets	Loads the smallest task-relevant workbench
Least-Action Router	Chooses the smallest sufficient route
Expansion Warrants	Requires explicit justification for broad context or compute
Compression Debt	Records downstream cost caused by bad compression
Saved-Work Certificates	Proves equivalent work can be reused
Canon Pressure	Surfaces recurring concepts for operator promotion
Pathwaves	Compresses successful workflow shapes for replay
Anti-Fluff	Keeps filler out of hot operational memory
Full-Scope Registry	Exposes 620 registered features across engine families

AtomSmasher has receipt-backed benchmarks, but ratio claims are corpus-specific. The cold ledger and source pointers remain authoritative.

Context Crystal currently has 5/5 held-out quality parity with a minimum `1422.901x` operational context ratio across a 7,056,795-byte corpus. This workload-specific held-out result is never substituted for a live-turn ratio.

5. Hermes And Agent Features

Current Orange Hermes surface

- bounded leases;
- LOOM gates;
- approval and override auditing;
- filesystem, process, browser, and visual adapters;
- Misfit second opinion;
- execution receipts.

Official Hermes Agent product integration

Configured profiles:

- Navigator;
- Builder;
- Researcher;
- Reviewer;
- Visual;
- Misfit.

Configured product shape:

- one gateway owner;
- one Kanban dispatcher;
- one conductor and at most two specialists;
- filtered Orange Brain MCP tools;
- authenticated loopback OpenAI-compatible API;
- isolated profile state;
- no telemetry and no LAN bind.

The current integration manifest and static proof do not by themselves prove the product gateway, Kanban, and API are live. A fresh product preflight is required.

Fresh path status: Hermes Brain MCP delegation completed parent mediation, child execution, synthesis, receipts, and lease revocation in `11,409.53 ms`. That proves the harmless read contract; each mutating or long-running workflow keeps its own completion proof.

6. Visual And Creative Features

AE Eyes

- screenshot, image, and document analysis;

- persistent visual retrieval worker;
- browser DOM and accessibility inspection;
- structured visual results returned through Orange reports and receipts.

AE Eyes must prove actual visual input and structured output. A backend facade test alone is insufficient.

Captain Planet installed runtime routes

These have technically valid, independently decoded runtime artifacts, but studio quality is explicitly unassessed:

Role	Current model	Proven boundary
Image draft	FLUX.2 Klein 4B FP8	local weights and decoded image artifact
Video fallback	LTX-Video 2B 0.9.8 distilled	local weights and decoded motion artifact
Speech	Qwen3-TTS 1.7B CustomVoice	local model and non-silent WAV
Music	ACE-Step 1.5 turbo	local weights and non-silent WAV

Creative candidates

These remain `FUTURE_CANDIDATE` until installed, measured, and baked off:

- FLUX.2 Dev final/edit tier;
- LTX-2.5 modern audio/video tier;
- Hunyuan3D 2.1;
- PaddleOCR-VL 1.6;
- SeedVR2 restoration.

7. Current Awareness Features

- source intake and timestamping;
- primary-source preference;
- content hashes and immutable references;
- local suitability and risk classification;
- quarantine before adoption;
- incumbent-versus-candidate bakeoff;
- receipt-backed promotion.

Research packets inform decisions. They do not automatically install tools or rewrite canon.

8. Deployment Features

- two-computer preferred topology;
- single-computer supported architecture;
- immutable Orange deploy payload;
- pre-authored Hermes profiles and configuration templates;
- fixed manifest with recommended defaults and optional deselection;
- adoption of compatible installed Bun, Ollama, models, and tools;
- automatic download of missing selected components after explicit approval;
- all mutable runtime state under `%USERPROFILE%\OrangeBox-Data` ;
- hidden service expectation;
- authenticated Codexa rail;
- topology discovery requirement for public installs;
- resumable, checksum-verified model downloads as the installation contract;
- rollback-safe source and model updates;
- optional hosted frontier lease.

Generic public clean-install, updater, and arbitrary-hardware proof must be checked in ReadyForGit. This guide does not claim those gates are closed.

9. Optional Hosted Features

OpenAI 5.6, Codex, Claude, or another strongest hosted agent may be used for:

- guided installation;
- architecture and code review;
- frontier reasoning lease;
- difficult debugging;
- independent evidence review.

They are optional. Orange policy, memory, receipts, and deterministic execution remain local and cannot depend on a subscription.

Related Guides

- [Quick Start](#)
- [Operator Manual](#)
- [Technical Architecture](#)
- [LLM Operator Guide](#)
- [Model Installation Guide](#)